

A Computer Vision Framework for Multi-Class Player Detection and Tracking in Soccer Broadcast Footage

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Abstract

The use of artificial intelligence in soccer analytics is rapidly expanding, yet most professional tracking systems still rely on expensive multi-camera setups that limit accessibility for lower-resource teams. This paper investigates whether coordinate and event-level information can be reliably extracted from raw broadcast footage using a single-camera, computer vision-based pipeline. In this paper we develop a system that combines a YOLO object detection model, a keypoint pitch detector, and the ByteTrack tracking algorithm to identify and track players, referees, goalkeepers, and the ball across match frames. The model was trained on a custom dataset containing professional broadcast footage as well as additional gameplay recordings.

Results show that the pipeline achieves high performance in detecting players and officials, with strong precision, recall, and mAP scores across most classes. Ball detection remains the primary challenge due to its small size, rapid movement, and frequent occlusion. Despite this limitation, the findings demonstrate that AI can reliably extract player-level spatial information from a single broadcast camera.

This work highlights the potential for affordable, scalable computer vision systems in soccer analytics. By reducing the reliance on specialized hardware, the proposed approach opens the door for colleges, academies, and amateur clubs to adopt data-driven methods traditionally accessible only to professional teams.

Introduction

The topic of this paper is the application of computer vision in soccer. While the sport itself is known by different names—soccer in North America and football in most other parts of the world—the global conversation around the game is evolving rapidly, especially with North America set to host the 2026 World Cup. Beyond debates about terminology, new discussions are emerging around the role of artificial intelligence in soccer. Clubs that

adopted data analytics early have already demonstrated its competitive value, inspiring others to follow.

One of the most notable examples is Brighton & Hove Albion. Their analytics-driven recruitment strategy has produced multiple high-profile successes: Moisés Caicedo was purchased for £4 million and later sold for £115 million; Alexis Mac Allister joined for £6.9 million and was sold for £35 million; Kaoru Mitoma arrived for £2.6 million and is now valued above £50 million (Transfermarkt). These are not isolated cases but indicators of a systematic, data-informed process. Brighton's Technical Director, David Weir, highlighted this advantage when he said that the club is "fortunate to have access to all the information from every league in the world, which you'd never be able to cover on a scouting, subjective, eyes-on basis." (Kershaw) Analytics have contributed significantly to Brighton becoming one of the most profitable clubs in the Premier League during the period from 2019 to 2023 (Swiss Ramble).

This context demonstrates why the integration of data and AI is transformative. However, data does not appear by itself. As Weir implied, it is impossible for scouts to watch every player across every league and extract meaningful insights manually. Computer vision offers a scalable solution: it allows clubs to construct pipelines that automatically extract player, ball, and event information from raw match footage. This paper focuses on that foundational step.

The research problem addressed here is twofold. First, automatic data extraction begins with accurately detecting and tracking all key entities on the pitch: players, goalkeepers, referees, and the ball. The question is whether modern AI models can achieve this reliably. Second, soccer analytics often require coordinate and event data, yet many clubs lack GPS-based tracking systems. This raises another question: can coordinate data be captured reliably from broadcast or match footage using computer vision?

Although AI applications in sports are growing, computer vision in soccer remains relatively new, and rapid advances in machine learning make earlier studies quickly outdated. Many prior works rely on older architectures such as RNNs or Faster R-CNN. In contrast, newer real-time detection models like YOLO have demonstrated superior performance and efficiency but have not yet been widely evaluated in this specific context.

This paper contributes to the current body of research by applying an up-to-date methodology using a modern object detection model to evaluate how effectively players and other key entities can be detected in soccer footage. The approach involved splitting raw match video into individual frames and processing them using two models: an object detection model to identify players, referees, goalkeepers, and the ball, and a pitch

keypoint detection model to support future coordinate extraction. This paper focuses primarily on the object detection component.

The findings indicate that the model performs well in detecting players, referees, and goalkeepers, though detecting the ball remains challenging due to its small size, speed, and frequent occlusion.

The remainder of the paper is structured as follows: the Related Work section reviews prior research and industry applications; the Methodology section describes the models and processing pipeline; the Results section presents detection performance; the Discussion section interprets the findings, acknowledges limitations, and outlines future research directions; and the Conclusion summarizes the study's contributions.

Related Work

Research on applying computer vision to soccer has grown significantly in recent years, though the field is still in its early stages. Several major approaches have been explored to detect, track, and analyze players and events in match footage. One of the more comprehensive works is by Jonas Theiner et al., who experimented with Faster R-CNN, FootAndBall, and CenterTrack to detect and track players from broadcast video. Their methodology notably considered various camera angles, incorporating the dynamic nature of broadcast footage into their tracking pipeline. Another relevant contribution is by Markappa, who presented an extensive discussion on the evolution of the YOLO family of object detectors and highlighted the improvements across YOLO generations. However, the paper did not specifically analyze performance across different object classes, leaving room to explore how modern models behave on players, referees, and the ball.

In terms of foundational or influential work, both Theiner et al. and Markappa provide valuable insights. Theiner's work is influential for its attention to real-world broadcast conditions, while Markappa's overview of YOLO's development offers important context for understanding how modern detectors emerged. Still, because computer vision in soccer is a relatively new application area, the field lacks a large body of standardized, widely cited literature. Research produced even a few years ago can quickly become outdated as newer, more efficient architectures are introduced.

A significant challenge in this domain is the lack of publicly available datasets. Unlike fields such as pedestrian detection or autonomous driving—where large, labeled datasets exist—soccer video datasets are scarce. Many professional clubs maintain private internal datasets generated by in-house analytics teams, but these are rarely shared publicly. As a

result, academic research often relies on small, custom-created datasets, making replication and benchmarking more difficult.

Existing approaches provide several strengths. Theiner et al.'s work is valuable because it recognizes the complexities of multi-camera broadcast footage and accounts for camera transitions. Markappa's analysis contributes by clearly outlining the capabilities of different YOLO versions and how they improve upon earlier architectures. However, these works also have limitations. Theiner et al. rely on architectures such as Faster R-CNN that have since been surpassed by more modern models like YOLO in both speed and accuracy. Markappa's work, while useful as an overview, focuses primarily on comparing YOLO variants rather than applying them to optimize real-world performance in soccer-specific contexts.

Compared to prior research, this study builds upon the foundational ideas introduced by these works while advancing the methodology through the use of a modern YOLO model. My work examines the performance of YOLO not only as a general detector but specifically across individual classes—players, referees, goalkeepers, and the ball. Additionally, this study incorporates a keypoint pitch detection model capable of identifying 32 field landmarks, enabling future research on coordinate extraction and event mapping. This contribution is necessary because soccer-focused computer vision research remains limited. To the best of my knowledge, no existing work has applied the most recent YOLO models to soccer footage while evaluating class-specific performance in detail. By providing updated benchmarks and integrating pitch keypoint detection, this study helps move the conversation forward by establishing a stronger baseline for future research and practical applications in sports analytics.

Overall, the field is evolving rapidly. Advances in deep learning over the past decade—moving from RNN-based detectors to Faster R-CNN and now to real-time architectures like YOLO—reflect a broader trend toward models that are both faster and more accurate. As computer vision continues to improve, its adoption in soccer analytics is likely to grow, enabling more advanced tracking, event detection, and automated analysis tools.

Methodology

Overall Pipeline

The overall approach of this study consists of a multi-stage computer vision pipeline designed to detect, track, and analyze players, referees, goalkeepers, and the ball from broadcast soccer footage. Raw video was first decomposed into individual frames. Each frame was processed by two separate models: (1) an object detection model based on

YOLOv8s to identify players and other objects, and (2) a keypoint-based pitch detection model capable of locating 32 field landmarks. After object detection, detections were passed into ByteTrack to assign persistent unique IDs, enabling player tracking across consecutive frames. To separate players into their respective teams, I used embedding-based clustering. SigLib was used to extract embeddings from each player's cropped image, UMAP for dimensionality reduction, and KMeans to cluster the embeddings into two teams. This pipeline allowed not only detection and tracking but also team-level segmentation for downstream analysis.

Data

The dataset used in this study was a custom collection from Roboflow, which included annotated broadcast footage from the German Bundesliga. To increase diversity in camera angles, lighting, and skill levels, I supplemented this footage with personal match recordings from my NCAA Division I and UPSL matches. Combining professional and semi-professional footage allowed the model to generalize more effectively across different contexts.

Data Preprocessing

Before training, all raw videos were split into frames at a consistent frame rate. Each frame was resized and normalized according to YOLO's preprocessing standards. Bounding box annotations provided by Roboflow were converted into YOLO label format. Additional augmentations such as horizontal flipping, rotation, brightness changes, and random cropping were applied to increase robustness to broadcast variations. All images were converted to RGB and scaled to the chosen input size of 1280×1280 pixels to ensure finer granularity when detecting small objects such as the ball.

Model Architecture

The primary detection architecture used was the YOLOv8s model. Although training initially began with the YOLOv8x variant, hardware constraints—including GPU memory limitations—made the larger model impractical. YOLOv8s provided an optimal balance between computational efficiency and performance.

The model takes in raw RGB frames through an initial convolutional layer that converts the 3-channel image into 32 feature maps. The backbone of the network contains 127 hidden layers composed of:

- Convolutional Layers — These form the basis of visual feature extraction, detecting edges, textures, patterns, and shapes.

- Batch Normalization + Activation Functions — These stabilize intermediate activations and introduce non-linearity, allowing the model to learn complex relationships.
- C2f/CSP Blocks — Lightweight but expressive blocks designed for efficient deep feature extraction without unnecessary computational cost.
- Downsampling and Upsampling Layers — These enable multi-scale feature learning, allowing the model to detect large and small objects (e.g., full players vs. the ball).
- SPPF (Spatial Pyramid Pooling Fast) — This component captures global spatial context, improving the model's ability to reason across varying camera distances.
- Concatenation Layers — These fuse feature maps from different resolutions to enable holistic, multi-scale reasoning.
- Detection Head — The final layer outputs bounding boxes, class probabilities, and confidence scores.

Together, these components form an efficient end-to-end detection pipeline suited for real-time soccer analysis.

Training Procedure

To better detect small objects—specifically the ball—the input resolution was increased from the default 640×640 to 1280×1280 pixels. This higher resolution preserved finer spatial details and significantly improved ball visibility. A batch size of 6 was selected to fit GPU memory constraints and accelerate training. Training ran for 25 epochs to reduce computational cost while still achieving convergence. The optimizer used was AdamW, which is particularly effective for computer vision tasks because it decouples weight decay from gradient updates. This leads to more stable training, better generalization, and less overfitting—especially when using high-resolution inputs and relatively small datasets. During training, I monitored three primary loss components:

- Box Loss — Measures how accurately the predicted bounding boxes align with ground truth.
- Classification Loss — Measures how well the model classifies players, referees, goalkeepers, and the ball.
- Distribution Focal Loss (DFL) — Helps the model better localize objects by predicting a more precise distribution of bounding box edges.

These combined losses provided a comprehensive assessment of the model's learning progress.

Evaluation Metrics

Model performance was evaluated using precision, recall, mAP50, and mAP50–95, the standard metrics for object detection. Precision measured the proportion of correct detections out of all detections (i.e., low false positives). Recall measured how many real objects were successfully detected (i.e., low false negatives). mAP50 evaluated accuracy at an Intersection over Union (IoU) threshold of 0.5. This is the primary detection score and is widely used for comparing models. mAP50–95 calculated mean Average Precision across IoU thresholds from 0.5 to 0.95 in steps of 0.05, providing a more thorough understanding of detection quality at varying levels of strictness. These metrics collectively allowed for a detailed view of model performance across all object classes.

Original Contributions

A key original contribution of this work is the class-specific performance reporting. To the best of my knowledge, prior studies in soccer-focused computer vision do not break down detection metrics by class. This study evaluates class-level performance for players, referees, goalkeepers, and the ball, providing more fine-grained insights into model behavior. Additionally, integrating YOLOv8 with a pitch keypoint detector represents a modern, up-to-date methodological contribution to soccer analytics research.

Results

To evaluate the performance of the proposed detection and tracking pipeline, I report four standard object-detection metrics: Precision, Recall, mAP50, and mAP50–95. These metrics are widely used in computer vision because they assess different and complementary aspects of model performance. Precision measures the proportion of correct detections and indicates how well the model avoids false positives, which is particularly relevant in scenarios where objects such as cleats or heads may resemble the ball. Recall evaluates how many real objects were successfully detected, reflecting the model's ability to avoid missed detections. The mAP50 metric, computed at an IoU threshold of 0.5, provides a general measure of detection accuracy and is the most commonly reported score for YOLO-style detectors. Finally, mAP50–95 averages mAP across stricter IoU thresholds (0.5 to 0.95), offering a more rigorous evaluation of precise localization. Together, these metrics give a comprehensive understanding of detector reliability and bounding-box quality.

Quantitatively, the model demonstrates strong performance across most classes. For all objects combined, the model achieves a precision of 0.884, recall of 0.828, mAP50 of 0.871, and mAP50–95 of 0.583 across 1,174 total instances. Player detection is particularly

strong, with precision and recall near perfect (0.957 and 0.978, respectively) and an mAP50 of 0.993. Goalkeepers and referees also exhibit reliable detection, each with mAP50 values above 0.93. In contrast, ball detection is significantly more challenging. Although precision remains high at 0.914, recall drops to 0.511 and mAP50 to 0.616, indicating that while the model is usually correct when it identifies the ball, it frequently fails to detect it under difficult conditions. The large gap between player-related metrics and ball-related metrics reflects fundamental differences in object size, visibility, and occlusion patterns.

Direct comparison to prior work is limited by the lack of standardized public benchmarks that include the same set of soccer-specific classes. However, relative to performance reported in similar sports-analytics studies using YOLO-based architectures, my model performs competitively for players, referees, and goalkeepers, often matching or exceeding baseline mAP values. The primary area where performance lags—ball detection—is also well documented in existing literature, where the ball’s small size, motion blur, and frequent occlusion consistently degrade detector performance. Thus, the observed results align with known challenges in sports-tracking research.

Qualitatively, detection and tracking visualizations support the numerical findings. In frames with clear visibility, the model consistently identifies all players, assigns stable tracking IDs via ByteTrack, and correctly clusters players into their respective teams using UMAP embeddings and KMeans clustering. Success cases show robust tracking even during player movement, crowding, or partial overlaps. Failure cases predominantly involve the ball: instances where the ball is fully or partially hidden behind players, blurred during rapid movement, or similar in appearance to cleats or light-colored head regions often lead to missed detections or occasional false positives. These visual analyses reinforce the statistical observation that player detection is highly reliable, while ball detection is the primary source of error.

Several trends emerge from the results. The model generalizes well to broadcast-style footage, maintaining strong detection accuracy across varying camera angles, lighting conditions, and field layouts. The very high performance on players, referees, and goalkeepers suggests that the YOLOv8s architecture captures the relevant spatial features effectively, even at the smaller model size. However, performance on the ball demonstrates the limitations of standard 2D detection methods when applied to extremely small or occluded objects. The discrepancy between high precision and low recall for the ball indicates that the model understands the ball’s characteristics but often lacks sufficient visual evidence to confidently detect it. These trends highlight both the strengths and challenges inherent in broadcast soccer analysis.

Despite the overall strong performance, the results reveal several limitations. The most significant is the difficulty of accurately detecting the ball, particularly in crowded scenes or when the camera angle obscures the object. False positives sometimes occur when cleats or bald heads resemble the ball, and false negatives arise when the ball is occluded by players' bodies. Additionally, the reliance on higher-resolution inputs (1280×1280) improves ball visibility but increases computational cost, which may limit real-time applicability on lower-resource systems. These limitations suggest potential directions for future work, including multi-frame ball reasoning, temporal smoothing, or specialized ball-focused detection modules.

Discussion

The results of this work demonstrate that the proposed detection and tracking pipeline performs reliably in identifying players, referees, and goalkeepers, but several limitations remain that constrain its broader applicability. First, although ByteTrack assigns unique IDs to each player, these identifiers are randomly generated and do not correspond to jersey numbers or stable player identities across long sequences. When a player exits the frame and later re-enters, the model assigns a new ID, which interrupts continuous identity tracking. This presents a challenge for applications requiring persistent player-level statistics, such as monitoring individual performance, physical metrics, or tactical behavior. Additionally, because the model was trained primarily on broadcast-style footage, it generalizes poorly to alternative camera angles such as elevated tactical views, handheld sideline recordings, or drone perspectives. This angle sensitivity reflects both dataset limitations and the difficulty of training models to handle diverse geometric distortions in sports footage.

These limitations suggest several directions for future work. The ability to extract accurate coordinate data from video is foundational for building fully automated analytical systems in soccer. With high-quality positional data, researchers can derive event data, compute advanced team-level metrics, and generate player-specific statistics—capabilities traditionally reserved for professional clubs with expensive multi-camera setups. Expanding the training dataset to include footage captured from various angles would significantly improve model robustness and reduce its dependence on broadcast views. Furthermore, solving the player re-identification problem—ensuring that players maintain consistent identities even after leaving and re-entering the frame—would move the system closer to real-world usability. Techniques such as appearance-based re-identification networks, sequence modeling, or temporal feature aggregation could support this improvement.

Interpreting these results in the context of the original problem, the findings indicate that AI-based systems can reliably detect and track soccer players from standard broadcast video. This is a critical first step toward automating the extraction of meaningful analytics from matches. Unlike professional teams, which typically rely on 8–12 synchronized cameras to reconstruct player coordinates and events, this pipeline operates effectively using only a single camera. This represents a major advantage for colleges, academies, and amateur clubs that lack the budget or infrastructure for advanced tracking systems. By demonstrating that high-quality detections can be achieved with minimal hardware, this work lowers the barrier to entry for data-driven performance analysis and provides a foundation for more accessible, scalable soccer analytics tools.

Conclusion

This paper set out to address the core problem of extracting reliable coordinate data from raw match footage using only a single broadcast camera. Achieving this goal required developing a pipeline capable of detecting and tracking players, referees, goalkeepers, and the ball directly from video frames. To accomplish this, I employed a modern You Only Look Once (YOLO)–based object detection model combined with a tracking algorithm and a pitch keypoint detector, forming a complete system for identifying and localizing key individuals on the field.

The results demonstrate that artificial intelligence can reliably detect and track players, referees, and goalkeepers in broadcast footage, confirming that single-camera systems are a viable pathway toward affordable, scalable soccer analytics. The primary limitation lies in ball detection: because the ball is small, fast-moving, and frequently occluded, the model struggles to track it consistently. Nonetheless, the work provides strong evidence that AI-based pipelines can extract meaningful positional information from match footage without the costly multi-camera setups used in professional clubs.

These findings are significant for both the sports analytics research community and practical applications in soccer. For professional clubs, this work helps validate the role of computer vision in enabling automated, data-driven analysis. For colleges, academies, and amateur teams—organizations that lack access to advanced tracking technologies—this approach offers a realistic and accessible path to integrating AI into scouting and performance evaluation. By lowering the resource barriers to data collection, this work expands the potential user base for analytical tools and supports more equitable access to modern recruitment methods.

The main insight gained from this study is that reliable player detection and tracking can be achieved using current state-of-the-art models and a single camera, showing that the foundational components of an automated analytical pipeline are achievable with modest hardware. However, important limitations remain. The model was trained primarily on broadcast footage, which restricts its generalizability to other camera angles or lower-quality recordings. Future work should include diversifying the training data, improving ball detection, and developing more robust re-identification methods to maintain consistent player identities across longer sequences.

In closing, this paper demonstrates that AI-driven soccer analytics are both feasible and increasingly accessible. By combining modern computer vision techniques with practical, real-world footage, this work contributes to the growing body of research at the intersection of sports and artificial intelligence and moves the field toward scalable, automated, and widely usable performance analysis tools.

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